The empirical recurrence for OEIS A297587: recovering a conjecture that is not written down

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Abstract

OEIS A297587 records "Empirical recurrence of order 94", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 259$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 94, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 94 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A297587 is "Number of 6Xn 0..1 arrays with every 1 horizontally, diagonally or antidiagonally adjacent to 1 or 4 neighboring 1s.". It has offset 1 and begins

$$1, 85, 1243, 4428, 31171, 395011, 3355439, 21941552, \ \ldots$$

The entry states:

> Empirical recurrence of order 94 (see link above)

As of the "Last modified" line on the live entry (Jan 01 2018 11:16:32, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 4096$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 259$ of the 4096, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 259$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 518$ exact terms of the model returns order 94 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{94} c_i\, a(n-i),$$

c_1	3	c_{25}	137108591465	c_{49}	867681222081534	c_{73}	−222471581806
c_2	20	c_{26}	−994236847874	c_{50}	37581719035671	c_{74}	−12201715916826
c_3	172	c_{27}	1955225277334	c_{51}	−918871763233842	c_{75}	−8696249801054
c_4	1032	c_{28}	103654059302	c_{52}	−1330149976309823	c_{76}	−6948743773369
c_5	−4118	c_{29}	3707670663704	c_{53}	−454651944107685	c_{77}	4469444939603
c_6	−26121	c_{30}	−6162368985712	c_{54}	579706856082091	c_{78}	6567005248160
c_7	−72630	c_{31}	−3850665298877	c_{55}	1308172182829870	c_{79}	373278811872
c_8	−4175	c_{32}	−11650754883238	c_{56}	865130343628788	c_{80}	−669619001958
c_9	1717096	c_{33}	8564145041322	c_{57}	−74813749268167	c_{81}	−1169298730770
c_{10}	3907660	c_{34}	19409731789953	c_{58}	−631140917322574	c_{82}	−611615415166
c_{11}	−2878686	c_{35}	20182825335421	c_{59}	−685084586604634	c_{83}	63153376925
c_{12}	−26795520	c_{36}	8427264234011	c_{60}	−203560877434736	c_{84}	142643484698
c_{13}	−90783249	c_{37}	−40402120495144	c_{61}	184508373826857	c_{85}	103407324406
c_{14}	−60118989	c_{38}	−59261520686820	c_{62}	282252816777909	c_{86}	10485588578
c_{15}	341526095	c_{39}	−59138864478317	c_{63}	86256773415883	c_{87}	−27454609736
c_{16}	1185388570	c_{40}	25556248826528	c_{64}	−47479115777437	c_{88}	−10397496648
c_{17}	1272072813	c_{41}	103653086261304	c_{65}	−25241496866788	c_{89}	1516733600
c_{18}	−781231796	c_{42}	272487350275679	c_{66}	−7143122180029	c_{90}	1216106560
c_{19}	−11461492942	c_{43}	131522388694897	c_{67}	−5491175571451	c_{91}	234699648
c_{20}	−12221462624	c_{44}	−140054356084112	c_{68}	−22670135943976	c_{92}	82361856
c_{21}	8379670918	c_{45}	−482808650971833	c_{69}	−35692089421354	c_{93}	29675520
c_{22}	29584702099	c_{46}	−474561023015084	c_{70}	6191115384955	c_{94}	3430400
c_{23}	148134190457	c_{47}	65729013442431	c_{71}	32834912900344		
c_{24}	−287266062881	c_{48}	800841699219646	c_{72}	26436739822658		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 94, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $94 + 4$ terms removed returns order 94 as well, so $\deg q_{\min} = 94 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. ☐

4 Verification

Three checks.

First, the walk model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $94+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 94 that this sequence satisfies — and that family turns out to have exactly one member.

References

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